

Generation Interconnection Combined Feasibility / System Impact Study Report

For Queue Project Q18-0997

Speed 138 kV
November 23, 2019

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q18-0997. The Interconnection Customer proposes a 108.3 MW battery energy storage facility interconnecting at the Speed 138 kV substation in the Kentucky area. The request was received on May 16, 2018. The study evaluated system intact and single contingency conditions using the regional planning case, including 346 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 27 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$24,956,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

1. Facility: Silver Bay Power to Speed 138 kV
Contingency: 56Th Street to Oats Reit 138 kV
Loading: 208.9% (Rating: 212 MVA)
DFAX: 100.0%
Pre-project loading: 157.7%
2. Facility: Narod to Barnham 138 kV
Contingency: Speed to Silver Bay Power 138 kV
Loading: 196.5% (Rating: 191 MVA)
DFAX: 100.0%
Pre-project loading: 139.9%
3. Facility: Gaga to Barnham 138 kV Line
Contingency: Silver Bay Power to Speed 138 kV
Loading: 187.6% (Rating: 190 MVA)
DFAX: 100.0%
Pre-project loading: 130.7%
4. Facility: Narod to Brockport 138 kV
Contingency: Silver Bay Power Substation to Speed 138 kV
Loading: 184.7% (Rating: 221 MVA)
DFAX: 100.0%
Pre-project loading: 135.8%
5. Facility: Santa Monia to 56Th Street 138 kV
Contingency: Speed to Silver Bay Power 138 kV
Loading: 179.8% (Rating: 228 MVA)
DFAX: 100.0%
Pre-project loading: 132.2%

6. Facility: Oates Reit to 56Th St 138 kV
Contingency: Speed to Silver Bay Power 138 kV Line
Loading: 179.2% (Rating: 247 MVA)
DFAX: 100.0%
Pre-project loading: 135.3%
7. Facility: Hambrick to Santa Monica 138 kV
Contingency: Silver Bay Power to Speed 138 kV line
Loading: 178.7% (Rating: 218 MVA)
DFAX: 100.0%
Pre-project loading: 129.1%
8. Facility: Speed to Gaga 138 kV line
Contingency: Speed Substation to Silver Bay Power 138 kV Line
Loading: 162.2% (Rating: 209 MVA)
DFAX: 100.0%
Pre-project loading: 110.2%
9. Facility: Minco to Barrett 230 kV
Contingency: Norwich to Barrett 500 kV
Loading: 111.0% (Rating: 620 MVA)
DFAX: 8.1%
Pre-project loading: 109.6%
10. Facility: Toledo Bend to Wharton 230 kV
Contingency: Barrett to Norwch 500 kV
Loading: 110.5% (Rating: 702 MVA)
DFAX: 8.8%
Pre-project loading: 109.1%
11. Facility: Santa Rosa Beach to Tule 138 kV
Contingency: Davisville to N Appleton 138 kV
Loading: 109.3% (Rating: 239 MVA)
DFAX: 5.9%
Pre-project loading: 106.6%
12. Facility: Valemount Substation to Tristate 345 kV
Contingency: Gavin Power Llc to Sporn 230 kV
Loading: 109.3% (Rating: 1,168 MVA)
DFAX: 8.6%
Pre-project loading: 108.5%
13. Facility: W Mount Vernon to Santa Rosa Beach 138 kV
Contingency: Davisville to N Appleton 138 kV
Loading: 108.4% (Rating: 254 MVA)
DFAX: 5.9%
Pre-project loading: 105.9%
14. Facility: West Mount Vernon to Nsp 138 kV
Contingency: N Appleton to Davisville 138 kV
Loading: 107.7% (Rating: 270 MVA)
DFAX: 5.9%
Pre-project loading: 105.3%
15. Facility: Nsp Substation to Wissota 138 kV line
Contingency: Davisville to North Appleon 138 kV
Loading: 107.0% (Rating: 285 MVA)
DFAX: 5.9%
Pre-project loading: 104.8%
16. Facility: 15Th Avenue S to Lucent 230 kV Line
Contingency: Omni-Hackberry to Norwich 500 kV

Loading: 105.5% (Rating: 921 MVA)
DFAX: 5.3%
Pre-project loading: 104.9%

17. Facility: Rancho Carmel to Russellville E 138 kV line

Contingency: Albion Rd to Platte City 138 kV
Loading: 105.4% (Rating: 342 MVA)
DFAX: 7.5%
Pre-project loading: 103.0%

18. Facility: Norwich to 15Th Avenue South 230 kV L

Contingency: Omni-Hackberry to Norwich 500 kV
Loading: 105.2% (Rating: 939 MVA)
DFAX: 5.3%
Pre-project loading: 104.5%

19. Facility: Bay Pt 230/1kV XFMR

Contingency: Barrett to Wolfe 500 kV
Loading: 104.2% (Rating: 413 MVA)
DFAX: 10.5%
Pre-project loading: 101.5%

20. Facility: Berry St 345 kV/138 kV TX

Contingency: Omni-Hackberry to Norwich 500 kV
Loading: 103.3% (Rating: 547 MVA)
DFAX: 11.2%
Pre-project loading: 101.0%

21. Facility: Ironman to Goldsboro 230 kV

Contingency: Norwich to Omni-Hackberry 500 kV
Loading: 102.3% (Rating: 536 MVA)
DFAX: 6.7%
Pre-project loading: 101.0%

22. Facility: Valemount to Pence 345 kV Line

Contingency: Sporn to Gavin Power Llc 230 kV
Loading: 102.2% (Rating: 1,069 MVA)
DFAX: 8.3%
Pre-project loading: 101.4%

23. Facility: Union to Garden Villas 138 kV

Contingency: Union Substation to Desoto Pleasant Run 138 kV
Loading: 101.6% (Rating: 194 MVA)
DFAX: 5.5%
Pre-project loading: 98.5%

24. Facility: Baraga Vlg 500/230 kV Transformer Bank

Contingency: Beaver to Trego 500 kV
Loading: 101.5% (Rating: 1,689 MVA)
DFAX: 8.7%
Pre-project loading: 100.9%

25. Facility: Madeline to E Methuen 230 kV

Contingency: Gray'S Flat to E Methuen 230 kV Line
Loading: 100.7% (Rating: 1,597 MVA)
DFAX: 8.7%
Pre-project loading: 100.1%

26. Facility: Berry St to Loveland 138 kV

Contingency: Berry St to Northfield 345 kV
Loading: 100.1% (Rating: 328 MVA)

DFAX: 7.0%
Pre-project loading: 97.8%

27. Facility: Loveland to Ojo 138 kV
Contingency: Berry St to Northfield 345 kV
Loading: 100.0% (Rating: 332 MVA)
DFAX: 7.0%
Pre-project loading: 97.7%

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Silver Bay Power to Speed 138 kV	Rebuild line with higher-capacity conductor	\$23,181,000	34% (\$7,850,000)
Narod to Barnham 138 kV	Rebuild line with higher-capacity conductor	\$14,551,000	34% (\$4,928,000)
Gaga to Barnham 138 kV Line	Rebuild line with higher-capacity conductor	\$1,516,000	34% (\$513,000)
Narod to Brockport 138 kV	Rebuild line with higher-capacity conductor	\$3,361,000	34% (\$1,138,000)
56Th Street to Santa Monica 138 kV	Rebuild line with higher-capacity conductor	\$3,830,000	34% (\$1,297,000)
Oates Reit to 56Th St 138 kV	Rebuild line with higher-capacity conductor	\$2,993,000	34% (\$1,014,000)
Hambrick to Santa Monica 138 kV	Rebuild line with higher-capacity conductor	\$4,005,000	34% (\$1,356,000)
Speed to Gaga 138 kV line	Rebuild line with higher-capacity conductor	\$6,377,000	34% (\$2,160,000)
Minco to Barrett 230 kV	Rebuild line with higher-capacity conductor	\$9,982,000	2% (\$253,000)
Toledo Bend to Wharton 230 kV	Rebuild line with higher-capacity conductor	\$19,891,000	2% (\$471,000)
Santa Rosa Beach to Tule Substation 138 kV	Reconductor line	\$1,381,000	4% (\$55,000)
Valemount Substation to Tristate 345 kV	Reconductor line	\$23,569,000	1% (\$270,000)
W Mount Vernon to Santa Rosa Beach 138 kV	Reconductor line	\$1,031,000	4% (\$41,000)
Nsp to W Mount Vernon 138 kV	Reconductor line	\$2,254,000	4% (\$89,000)
Wissota to Nsp 138 kV	Reconductor line	\$7,088,000	4% (\$281,000)
15Th Avenue S to Lucent 230 kV Line	Reconductor line	\$7,283,000	4% (\$261,000)
Russellville East to Rancho Carmel 138 kV	Reconductor line	\$2,778,000	4% (\$121,000)

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Norwich to 15Th Avenue S 230 kV Line	Reconductor line	\$5,645,000	4% (\$202,000)
Bay Pt 230/1kV XFMR	Replace transformer with higher-rated unit	\$9,004,000	3% (\$250,000)
Berry St 345 kV/138 kV TX	Replace transformer with higher-rated unit	\$12,386,000	4% (\$487,000)
Ironman to Goldsboro 230 kV	Reconductor line	\$21,296,000	2% (\$461,000)
Valemount to Pence 345 kV Line	Reconductor line	\$9,163,000	1% (\$109,000)
Union to Garden Villas 138 kV	Reconductor line	\$4,801,000	5% (\$234,000)
Baraga Vlg 500/230 kV Transformer Bank	Replace transformer with higher-rated unit	\$25,942,000	2% (\$616,000)
Madeline to East Methuen 230 kV	Reconductor line	\$11,735,000	2% (\$284,000)
Loveland to Berry St 138 kV	Reconductor line	\$3,790,000	4% (\$141,000)
Loveland to Ojo 138 kV line	Reconductor line	\$1,991,000	4% (\$74,000)
	Total allocated to project		\$24,956,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.